

Dynamical system with time step δ satisfies

$$x_{t+\delta} = [I + \delta A(t)] x_t.$$

Therefore,

$$\frac{x_{t+\delta} - x_t}{\delta} = A(t)x_t.$$

If $\delta \rightarrow 0$, then

$$\frac{dx_t}{dt} = A(t)x_t,$$

or

$$dx_t = A(t)x_t dt.$$

Let $C : [0, 2T] \rightarrow M_{n \times n}$ be a function taking value in the set of $n \times n$ matrices. Suppose that

$$A(t) = \begin{cases} C(t), & 0 \leq t \leq T; \\ -C(2T - t), & T \leq t \leq 2T, \end{cases}$$

then it is known that $x_{2T} = x_0$.